

Bounding higher-order ionosphere errors for the dual-frequency GPS user

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[1] Civil signals at L2 and L5 frequencies herald a new phase of Global Positioning System (GPS) performance. Dual-frequency users typically assume a first-order approximation of the ionosphere index of refraction, combining the GPS observables to eliminate most of the ranging delay, on the order of meters, introduced into the pseudoranges. This paper estimates the higher-order group and phase errors that occur from assuming the ordinary first-order dual-frequency ionosphere model using data from the Federal Aviation Administration's Wide Area Augmentation System (WAAS) network on a solar maximum quiet day and an extremely stormy day postsolar maximum. We find that during active periods, when ionospheric storms may introduce slant range delays at L1 as high as 100 m, the higher-order group errors in the L1–L2 or L1–L5 dual-frequency combination can be tens of centimeters. The group and phase errors are no longer equal and opposite, so these errors accumulate in carrier smoothing of the dual-frequency code observable. We show the errors in the carrier-smoothed code are due to higher-order group errors and, to a lesser extent, to higher-order phase rate errors. For many applications, this residual error is sufficiently small as to be neglected. However, such errors can impact geodetic applications as well as the error budgets of GPS Augmentation Systems providing Category III precision approach.

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1. Introduction

[2] The ionosphere is a weakly ionized plasma layer of the upper atmosphere from about 50 km upwards [Tascione, 1994]. As a dispersive medium, the ionosphere advances the phase and delays the code of the GPS ranging signals broadcast at L1 frequency $f_1 = 1575.42$ MHz and L2 frequency $f_2 = 1227.60$ MHz. For the currently broadcast L1 and L2 frequency signals, these result in errors in the code $P1$ and $P2$ and carrier phase measurements $L1$ and $L2$ made by the user:

$$P1 = \rho + I_P(f_1) \quad (1)$$

$$P2 = \rho + I_P(f_2) \quad (2)$$

$$L1 = \rho + I_L(f_1) + N_1 \lambda_1 \quad (3)$$

$$L2 = \rho + I_L(f_2) + N_2 \lambda_2 \quad (4)$$

[3] In these expressions, ρ includes the true range and all non-dispersive errors such as troposphere, satellite clock bias, and receiver clock bias. The ionosphere errors are denoted I , with a subscript L for the phase (or P for the group) error term. The carrier measurements at each frequency f_1 and f_2 are ambiguous by an integer N number of wavelengths λ . The receiver interfrequency bias, satellite hardware bias, multipath, and noise are not shown. The phase error I_L due to the signal passage through the ionosphere is:

$$I_L(f) = \int_{rx}^{sv} (n_L(f) - 1) dl \quad (5)$$

[4] By neglecting curvature and path differences for the L1 and L2 frequencies, we integrate (5) along the

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straight line of sight between the satellite *sv* and receiver *rx*.

[5] The phase index of refraction $n_L(f)$ for the extraordinary mode (i.e., right hand circularly polarized waves such as the GPS signals) propagating through the ionosphere, neglecting particle collisions, is given in *Bittencourt* [2004] by the Appleton-Hartree equation:

$$n_L^2 = 1 - \frac{X}{1 - \frac{Y^2 \sin^2 \theta}{2(1-X)} - \sqrt{\frac{Y^4 \sin^4 \theta}{4(1-X)^2} + Y^2 \cos^2 \theta}} \quad (6)$$

where

$$X = \frac{C_x N_e}{f^2}, \quad C_x = \frac{e^2}{4\pi^2 \epsilon_0 m_e} \quad (7)$$

$$Y = \frac{C_y B}{f}, \quad C_y = \frac{e}{2\pi m_e} \quad (8)$$

[6] In (6) and (8), B is the magnitude of the magnetic field and θ is the angle between the magnetic field direction and the wave propagation direction. The constants C_x and C_y include the electron charge e , electron mass m_e , and the permittivity of free space ϵ_0 .

[7] The Appleton-Hartree equation is unwieldy as a model for ranging, so simplifications are often made with a Taylor expansion about the signal frequency f and a series of approximations. Various approximations have been made in the literature to arrive at the following expression:

$$n_L = 1 - \frac{1}{2}X - \frac{1}{2}XY|\cos \theta| - \frac{1}{8}X^2 - \frac{1}{4}XY^2 \cos^2 \theta - \frac{1}{4}XY^2(\cos^2 \theta + \sin^2 \theta) \quad (9)$$

[8] The full equation (9) holds when $|\cos \theta| \gg \frac{Y \sin^2 \theta}{2(1-X)}$ [Bassiri and Hajj, 1993]. From (7) and (8) we see Y is inversely proportional to the wave frequency f , and X is inversely proportional to the square of the frequency. This implies that the second term goes as f^{-2} , the third term as f^{-3} , and the remaining terms as f^{-4} .

[9] The group index n_P is related to the phase index of refraction as [Jackson, 1999]:

$$n_P = n_L + f \frac{dn_L}{df} \quad (10)$$

$$= 1 + \frac{1}{2}X + XY \cos \theta$$

$$+ \frac{3}{8}X^2 + \frac{3}{4}XY^2 \cos^2 \theta + \frac{3}{4}XY^2 \quad (11)$$

where X and Y are given by (7) and (8), and θ is the angle between the magnetic field and the wave propagation direction.

[10] The first-order model of the ionosphere, the one most commonly used by dual-frequency users, includes the first two terms of (9) and (11) only [Misra and Enge, 2006]. *Brunner and Gu* [1991] retain the first four terms by order of magnitude considerations. *Tucker and Fannin* [1968] additionally retain the cosine terms by making a quasi-longitudinal approximation (i.e. $\theta = 0$). Both *Bassiri and Hajj* [1993] and *Hartmann and Leitinger* [1984] show the full equation (9). However, *Bassiri and Hajj* [1993] then reduce their model to the *Brunner and Gu* [1991] model for the purposes of analysis. As Klobuchar notes, the *Brunner and Gu* [1991] model error estimates are more accurate because they estimate curvature effects due to the signals not traveling along the straight line path, and moreover, not traveling on the same curved path. However, *Brunner and Gu*'s simulation requires more knowledge of the state of the ionosphere and geomagnetic field. *Bassiri and Hajj* [1993] neglect curvature as we do, treating the integration as taking place along the line of sight. They additionally consider the group index of refraction and the corresponding code errors, predicting them to be at the centimeter level. All these analyses used simulations of the ionospheric electron density to estimate the magnitudes of the higher-order ionosphere errors.

[11] More recent work has built on these derivations to consider the resultant positioning errors. *Kedar et al.* [2003] considered the second-order effect on positioning using dual-frequency data, showing a diurnal and seasonal variation in apparent position. *Hernandez-Pajares et al.* [2007] has considered the second-order effect on positioning for improvements to the International GNSS Survey (IGS). *Fritsche et al.* [2005] considered both the second- and third-order effects as they affect use of the International Terrestrial Reference Field (ITRF).

[12] This paper uses data from one of the most severe ionospheric storms of Solar Cycle 23 to estimate the higher-order ranging errors. Additionally, this paper addresses the effects of carrier smoothing on code observables, a widely used multipath mitigation technique in GPS receivers. Our focus on the range domain (as opposed to the position domain) allows us to consider the smoothing effects in receivers, a topic that has not otherwise been taken up in a significant way in the literature.

[13] Section 2 shows the expression for the phase and group errors when including the higher-order ionosphere terms and their appearance in the commonly formed "ionosphere-free" GPS observable. Sections 3 and 4 discuss approximations made to simplify these higher-order error terms. These simplifications allow us to estimate the higher-order terms from GPS observables,

as we show in section 5. The magnitudes of these higher-order terms are discussed in section 6. We show the effect of these higher-order terms on carrier smoothing of code observables in section 7 before making some closing remarks.

2. Higher-Order Errors

[14] In this section we show the expression for the higher-order ionosphere phase error and the error due to these terms for users of the standard first-order ionosphere model. Substitution of the phase index of refraction in (9) and the expressions for X and Y given in (7) and (8) into (5) gives the phase errors due to the ionosphere. To make the relationship between these terms and the powers of frequency more explicit, *Bassiri and Hajj* [1993] used variables q , s , and r , shown in equation (13), which we define in terms of the constants C_x and C_y in (14)–(19)

$$I_L(f) = - \int_{rx}^{sv} \frac{1}{2} X dl - \int_{rx}^{sv} \frac{1}{2} XY \cos \theta dl - \left[\int_{rx}^{sv} \frac{1}{8} X^2 dl + \int_{rx}^{sv} \frac{1}{4} XY^2 \cos^2 \theta dl + \int_{rx}^{sv} \frac{1}{4} XY^2 dl \right] \quad (12)$$

$$= -\frac{q}{f^2} - \frac{s}{2f^3} - \frac{r}{3f^4} \quad (13)$$

$$q = \frac{C_x}{2} \int_{rx}^{sv} N_e dl \quad (14)$$

$$s = C_x C_y \int_{rx}^{sv} N_e B \cos \theta dl \quad (15)$$

$$r = r_1 + r_2 + r_3 \quad (16)$$

$$r_1 = \frac{3C_x^2}{8} \int_{rx}^{sv} N_e^2 dl \quad (17)$$

$$r_2 = \frac{3C_x C_y^2}{4} \int_{rx}^{sv} N_e B^2 \cos^2 \theta dl \quad (18)$$

$$r_3 = \frac{3C_x C_y^2}{4} \int_{rx}^{sv} N_e B^2 dl \quad (19)$$

[15] A similar treatment for the group error to that of (12)–(13) gives the multiorder ionosphere as:

$$I_P(f) = \frac{q}{f^2} + \frac{s}{f^3} + \frac{r}{f^4} \quad (20)$$

[16] The terms containing s in (13) and (20) differ by a factor of -2 . Similarly, the r terms in those same equations differ by a factor of -3 .

[17] The dual-frequency GPS user typically assumes the ionosphere error is due only to the first-order (f^{-2}) term. The user forms an observable L^q that eliminates this term:

$$L^q = \frac{\gamma L1 - L2}{\gamma - 1} \quad (21)$$

$$\gamma = \frac{f_1^2}{f_2^2} \quad (22)$$

[18] The superscript q indicates that the first-order, or q term, is the highest-order error removed. Although the typical user assumes the higher-order errors are negligible, they remain in the first-order-free observable L^q . This can be seen by substituting (3), (4), (13), and (22) into (21):

$$L^q = \frac{\gamma}{\gamma - 1} \left(\rho - \frac{q}{f_1^2} - \frac{s}{2f_1^3} - \frac{r}{3f_1^4} \right) - \frac{1}{\gamma - 1} \left(\rho - \frac{q}{f_2^2} - \frac{s}{2f_2^3} - \frac{r}{3f_2^4} \right) + \frac{1}{\gamma - 1} (\gamma N_1 \lambda_1 - N_2 \lambda_2) \quad (23)$$

$$= \rho + \frac{s}{2f_1 f_2 (f_1 + f_2)} + \frac{r}{3f_1^2 f_2^2} + \beta \quad (24)$$

[19] In this expression, the integer ambiguities $N_1 \lambda_1$ and $N_2 \lambda_2$ have been absorbed into bias β .

[20] The corresponding form of the code measurement that is ionosphere-free to first order is:

$$P^q = \frac{\gamma P1 - P2}{\gamma - 1} \quad (25)$$

$$= \rho - \frac{s}{f_1 f_2 (f_1 + f_2)} - \frac{r}{f_1^2 f_2^2} \quad (26)$$

[21] *Klobuchar* [1996] comments that both *Brunner and Gu* [1991] and *Bassiri and Hajj* [1993] predict that

the higher-order phase errors (i.e. the s and r terms in (23)) would be on the order of 0.1% of the first-order error $\frac{q}{f^2}$ term via simulation. Since the typical ionosphere delay at L1 frequency is on the order of 10 m, this would translate to higher-order errors at the centimeter level.

[22] Our analysis will use data and some conservative approximations to isolate the effect that the higher-order terms s , r_1 , r_2 , and r_3 will have for a user forming the combination of the observables that are first-order-ionosphere free (21), which removes only the q term.

3. Magnetic Field Model

[23] In order to estimate the higher-order errors using the dual-frequency GPS observables $P1$, $P2$, $L1$, and $L2$, we make a number of simplifications to (15)–(19). For s in (15), r_2 in (18), and r_3 in (19), we need information about the magnetic field within the ionosphere. For this we assume the International Geomagnetic Reference Field IGRF-10 [Maus and Macmillan, 2005]. IGRF-10 is a spherical harmonic multipole expansion of the field, with 10th generation updated estimates of the coefficients that were used by Brunner and Gu [1991] and Bassiri and Hajj [1993]. Tsyganenko [2005] has made an implementation of the IGRF model publicly available as Fortran code, which we adapted for use in Matlab.

[24] In the s , r_2 , and r_3 ionosphere error terms, the magnitude B of the field must be integrated against the electron density N_e at each point along the raypath. Since we use real GPS data at L1 and L2 frequencies during ionospherically active times, we do not assume a Chapman function for the ionosphere. Therefore, we will not integrate the product of N_e and B . Instead, we use the magnitude B and the angle between field line and raypath θ at the point where the line of sight reaches 350 km altitude as the average values. The intersection of the raypath with a surface at 350 km altitude is known as the ionosphere pierce point, (IPP). This allows us to pull the B and $\cos \theta$ terms out of the integrals in (15), (18), and (19).

$$s \leq s' \int_{rx}^{sv} N_e dl, \quad s' = C_x C_y (B \cos \theta)|_{IPP} \quad (27)$$

$$r_2 \leq r'_2 \int_{rx}^{sv} N_e dl, \quad r'_2 = \frac{3C_x C_y^2}{4} (B \cos \theta)^2|_{IPP} \quad (28)$$

$$r_3 \leq r'_3 \int_{rx}^{sv} N_e dl, \quad r'_3 = \frac{3C_x C_y^2}{4} B^2|_{IPP} \quad (29)$$

[25] This average value is conservative since the geomagnetic field is reduced by currents induced in the

magnetosphere during ionospherically stormy times, as evidenced by negative D_{st} geomagnetic storm index [Campbell, 1996]. The use of the field value at the IPP is also conservative since evaluation at 350 km altitude treats the bulk of the electrons as being at that altitude, when in fact storm time plasma enhancements have been observed to occur at higher altitudes [Mannucci et al., 2005], where the magnetic field is weaker.

4. Simplification of N_e

[26] The r_1 term defined in (17) integrates the square of the electron density N_e along the line of sight. To estimate the r_1 term we use a technique introduced by Hartmann and Leitinger [1984]. We approximate the integral in (17) as follows:

$$\int_{rx}^{sv} N_e^2 dl \approx N_m \eta \int_{rx}^{sv} N_e dl \quad (30)$$

[27] In this expression N_m is the maximum density in the vertical density profile and η is a shape factor. In the literature, since Chapman functions with specified N_m were assumed, it was found that $\eta = 0.66$ nearly independently of elevation.

[28] Rather than use a Chapman function, we make an assumption that is both conservative and consistent with our data. We assume the ionosphere is a shell of thickness 100 km. Then we take our dual-frequency measurements of the first-order slant delay $\tilde{I}_s$, and convert them to equivalent vertical measurements $\tilde{I}_v$ using the obliquity factor M [Hansen, 2002]:

$$\tilde{I}_v = \frac{\tilde{I}_s}{M(el)} \quad (31)$$

$$M(el) = \frac{1}{\cos \left(\arcsin \left(\frac{R_E \cos(el)}{R_E + h_{iono}} \right) \right)} \quad (32)$$

[29] The obliquity factor M is a function of the elevation el to one satellite. The height of the ionosphere h_{iono} is fixed at 350 km, and R_E is the radius of the Earth. Then N_m is given by:

$$N_m = \frac{\tilde{I}_v}{\tau} = \frac{\tilde{I}_v}{100 \text{ km}} \quad (33)$$

[30] In other words, we compute the equivalent vertical first-order delay from the slant, and then spread the electrons causing that delay evenly within a shell of

thickness $\tau = 100$ km. Our model of electron density in equation (30) implies that $\eta = 1$.

[31] Compared to the International Reference Ionosphere (IRI) model described by *Bilitza* [2001], in which the vast majority of electrons are typically within a band of 150–200 km thickness, a narrow shell of 100 km thickness gives a higher value of N_m . Therefore, this assumption is conservative.

[32] A cartoon, not to scale, depicting this simplification is shown in Figure 1. It illustrates the variation of the electron density with height. An IRI or other Chapman-based model (black curve) increases to a peak and tapers off with altitude. A set of shells of varying thickness are shown as rectangles. Each one contains the same area (i.e. the same total electron content, TEC). The thinnest rectangle altitudinally represents a shell of thickness 100 km. This maximum density N_m associated with this choice of vertical profile is both higher than an IRI or other Chapman-based function would with the same TEC. It also gives a higher value of η than a Chapman function would. Therefore, it can be used to bound the r_1 error:

$$r_1 \leq r'_1 \int_{rx}^{sv} N_e dl, \quad r'_1 = \frac{3C_x^2}{8} N_m \eta \quad (34)$$

[33] Having redefined s , r_1 , r_2 , and r_3 with approximations s' , r'_1 , r'_2 , and r'_3 , we define q' and r' for notational consistency:

$$q = q' \int_{rx}^{sv} N_e dl, \quad q' = \frac{C_x}{2} \quad (35)$$

$$r \leq r' \int_{rx}^{sv} N_e dl, \quad r' = r'_1 + r'_2 + r'_3 \quad (36)$$

5. Higher-Order Errors From Data

[34] The goal of this section is to demonstrate how to extract the higher-order errors from differences of the carrier measurements $L1 - L2$. This is done because the data analyzed in section 6 are not the individual observables $L1$ and $L2$, but their difference.

[35] We are given the dual-frequency observable:

$$\tilde{I}_s = \frac{L1 - L2}{\gamma - 1} \quad (37)$$

[36] The constant γ is defined in (22). Substitute the expression for $I_L(f)$ of (13) into the definitions of $L1$ and $L2$ in (3) and (4) to get, assuming a procedure such as the

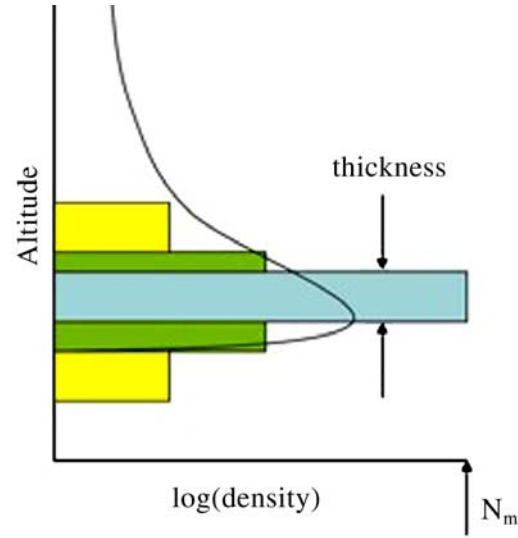


Figure 1. Defining an equivalent slab thickness τ and peak density N_m for the ionosphere.

one to be discussed in section 6 has been applied beforehand to eliminate integer ambiguity and hardware bias terms:

$$\begin{aligned} \tilde{I}_s &= \frac{1}{\gamma - 1} \left(\rho - \frac{q}{f_1^2} - \frac{s}{2f_1^3} - \frac{r}{3f_1^4} \right) \\ &\quad + \frac{1}{\gamma - 1} \left(-\rho + \frac{q}{f_2^2} + \frac{s}{2f_2^3} + \frac{r}{3f_2^4} \right) \\ &= \frac{q}{f_1^2} + \frac{s(\gamma^{3/2} - 1)}{2f_1^3(\gamma - 1)} + \frac{r(\gamma + 1)}{3f_1^4} \end{aligned} \quad (38)$$

[37] It can be seen that:

$$\begin{aligned} \frac{q}{f_1^2} &\leq \frac{q}{f_1^2} + \frac{s(\gamma^{3/2} - 1)}{2f_1^3(\gamma - 1)} + \frac{r(\gamma + 1)}{3f_1^4} \\ &\leq \tilde{I}_s \end{aligned} \quad (39)$$

$$\int_{rx}^{sv} N_e dl \leq \frac{2\tilde{I}_s f_1^2}{C_x} \quad (40)$$

[38] This shows that the integral appearing in equations (27)–(29) and (34) can be conservatively estimated from the measurements $\tilde{I}_s$ by dividing $\tilde{I}_s$ by the constants in the left-hand side of equation (39). To estimate the difference between the left and right sides of the inequality in equation (39), we substitute in our approximations of q , s , and r involving q' , s' , and r' . We find that the

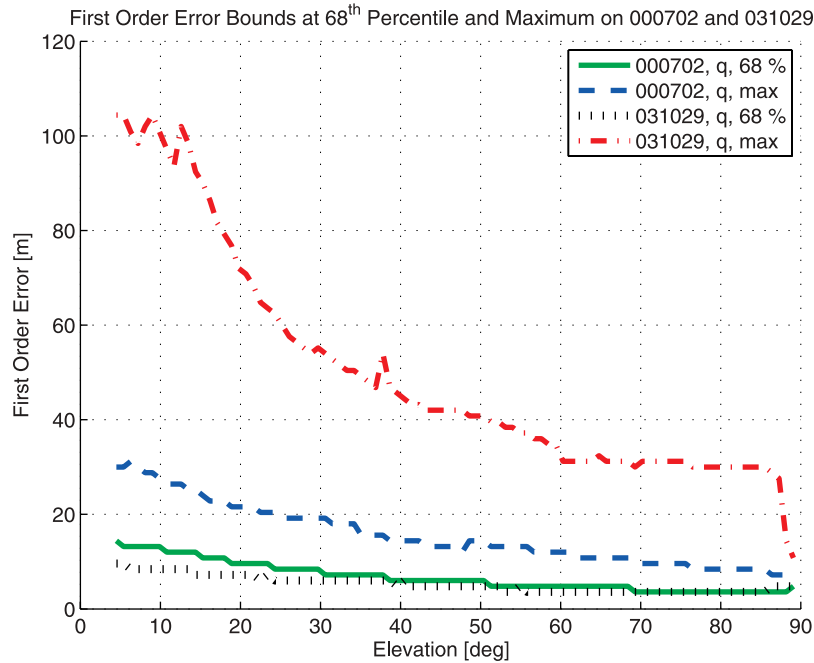


Figure 2. Group P error bounds versus elevation for quiet day 2 July 2000 and active day 29 October 2003. First-order q term $\sim Ne/f^2$ at 68th percentile and maximum value. Y-axis error scale 0 – 120 m. Phase error is equal and opposite.

second and third terms on the right hand side of equation (39) are 0.2% of the first term based on the data used. Based on this percentage we find that, using equation (40), we can bound the higher-order phase and group errors in (24) and (26). In the next section, we consider only the magnitudes of these errors, disregarding the sign.

[39] The higher-order errors for the L1–L5 dual-frequency user can be estimated by replacing f_2 with $f_5 = 1176.45$ MHz in (24) and (26). This is equivalent to scaling the second-order errors obtained in (24) and (26) by 1.063. Similarly, the third-order L1–L2 estimates may be increased by a factor of 8.9% to estimate the errors for the L1–L5 user.

6. Data

[40] Given the observable $\tilde{I}_s$, we now have a method to estimate each of the error terms that users of the first-order-iono-free expression L^q or P^q will suffer. The FAA's Wide Area Augmentation System (WAAS) network provides dual-frequency measurements, known as “supertruth” data. Each WAAS reference station has three receivers that collect measurements P_1 , P_2 , L_1 , and L_2 as given in (1)–(4). WAAS supertruth data consists of postprocessed dual-frequency measurements of the first-order ionosphere delay $\tilde{I}_s$ formed as given in (37).

[41] These dual-frequency carrier measurements are leveled to the code measurements to remove the integer ambiguities $n_1\lambda_1$ and $n_2\lambda_2$, and the interfrequency biases (not shown in the equations but listed in Hansen [2002]) are estimated by Komjathy [2005]. A voting system among the three receivers at each station requires consistency between the receivers to within tight bounds before the data are considered qualified to output [Komjathy et al., 2007]. From these measurements use equation (40) to overbound the integral $\int_{rx}^{sv} N_e dl$ and then substitute these values into (27)–(29) and (34)–(35).

[42] We consider one ionospherically quiet day, 2 July 2000, and one extremely active day, 29 October 2003. The magnitude of the first-order errors, due to the q term in equations (13) and (20), are plotted as a function of elevation angle in Figure 2. A histogram of the estimated error magnitude has been generated over all satellite-receiver pairs for the given day whose elevation angle falls within a given range. The 68th percentile, which represents one standard deviation for a normal distribution, is plotted in solid (quiet) and dotted (active). The 68th percentile values are comparable at all elevations. The maxima at each elevation are plotted in dashed (quiet) and dot-dashed (active) lines. On the quiet day, first-order errors ranged from as much as 10 m at high elevation to 30 m at low elevation. On the active day, first-order

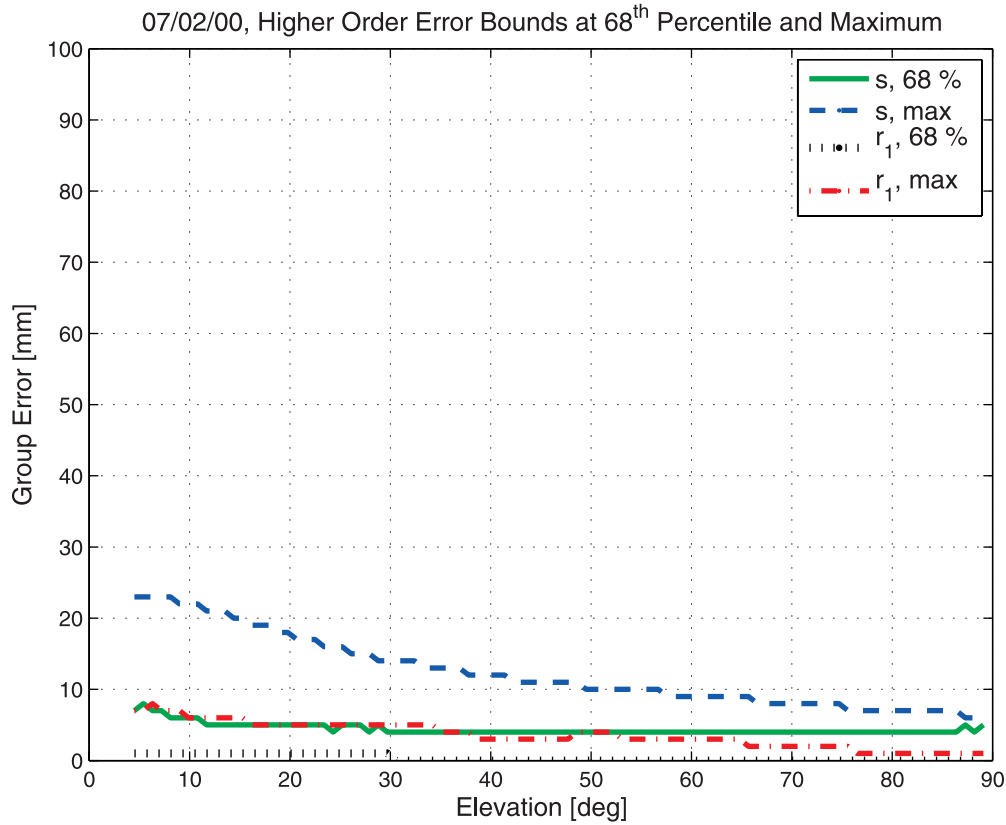


Figure 3. Group P error bounds versus elevation. Second-order $s \sim N_e B \cos \theta / f^3$ and third-order $r_1 \sim N_e^2 / f^4$ at 68th percentile and maximum value. Quiet day. Y-axis error scale: 0 – 100 mm.

errors were observed to be 30 m at high elevation up to 110 m at low elevation. In both cases the errors increase with decreasing elevation since low elevation lines-of-sight pass through more ionosphere to increase the total electron content along the slant path length.

[43] For the quiet day and active day, respectively, Figures 3 and 4 show the maximum and standard deviation of the second-order s term and third-order r_1 term as a function of elevation. The 68th percentile and maximum bounds on the group error due to the second-order s term are shown with solid and dashed lines, respectively. The errors due to the r_1 part of the third-order term in (26) are in dotted (68th) and dot-dashed (maximum) lines. On both figures, the y-axis scale ranges from 0 to 100 mm. The second-order phase errors are a factor of 2 lower in magnitude than the corresponding group errors shown. The phase errors due to r_1 are three times smaller than the analogous group errors.

[44] On both days the 68th percentile values of the errors are only a few millimeters, but the tails of the histograms are wider for the active day than the quiet day and are non-Gaussian. On the quiet day (Figure 3)

the highest second-order group errors (dashed line) are 25 mm at low elevation, whereas on the active day (on Figure 4) they are as much as 85 mm. On a quiet day the third-order errors due to the r_1 term (blue lines) proportional to N_e^2 are less than 10 mm even at low elevation (dot-dashed lines). However, on the active day the errors reach nearly 100 mm at low elevation. On the stormy day, the magnitude of the r_1 error (dot-dashed) is comparable to those of the s term errors (dashed).

[45] By contrast, the magnitude of the r_2 and r_3 group error terms is much much smaller, as is seen on Figure 5 for the quiet day and Figure 6 for the active day. Each plot contains the 68th percentile and maximum error as a function of elevation. The 68th percentile (solid) and maximum bounds (dashed) for r_2 and the 68th percentile (dotted) and maximum (dot-dashed) for r_3 are shown. On these plots, the y-axis scale ranges up to only 0.15 mm. The phase delay errors are three times smaller than the group error magnitudes shown in these plots. In both the group and phase cases, the r_2 and r_3 errors are submillimeter. For this reason, we find that *Brunner and Gu* [1991] and *Bassiri and Hajj* [1993] were justified in

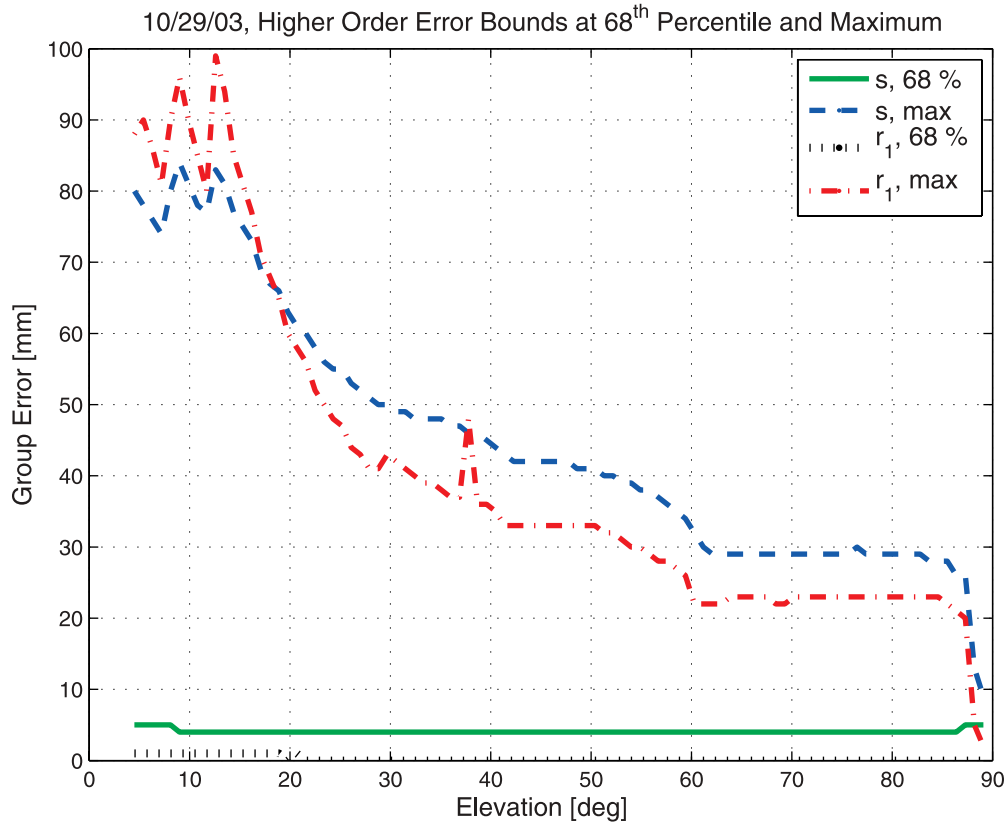


Figure 4. Group P error bounds versus elevation. Second-order $s \sim N_e B \cos \theta / f^3$ and third-order $r_1 \sim N_e^2 / f^4$ at 68th percentile and maximum value. Stormy day. Y-axis scale: 0 – 100 mm.

keeping only the q , s , and r_1 terms in their model of the index of refraction.

[46] Having estimated the magnitude of each of the higher-order errors, up to the fourth power in signal frequency, we discard the submillimeter terms and adopt the *Bassiri and Hajj* [1993] model for all subsequent analysis. We plot the total error arising in the standard dual-frequency model due to the q , s , and r_1 terms in phase equation (24) and group equation (26), for the quiet day and active day, as a function of elevation. These are shown in Figure 7. The scale of the y-axis reaches up to 200 mm.

[47] On an active day, the maximum higher-order phase errors can reach up to 80 mm at low elevation, and a fraction of that on quiet days. These figures indicate that, based on actual data and the assumptions we have made, on an active day the higher-order group errors can reach as high as 180 mm at low elevation. On the quiet day the errors are only about 20% those of the active day.

[48] One of the primary conclusions we draw from comparing the phase errors to the group errors in Figure 7

is that, even though they have opposite signs in the equations, they are no longer equal in magnitude. This will have an effect in smoothing of the first-order-free code observable with the first-order-free carrier observable.

7. Carrier Smoothing Errors

[49] In this section we examine how the higher-order group and phase errors affect carrier smoothing of the code observable that is first-order-ionosphere-free (i.e. the q term has been eliminated). For this analysis we reproduce the expressions for the carrier and code first-order-ionosphere-free observables that were shown in (24) and (26):

$$L^q = \rho + \frac{s}{2f_1 f_2 (f_1 + f_2)} + \frac{r}{3f_1^2 f_2^2} + \beta \quad (41)$$

$$P^q = \rho - \frac{s}{f_1 f_2 (f_1 + f_2)} - \frac{r}{f_1^2 f_2^2} \quad (42)$$

[50] In these expressions the true range and non-dispersive range terms are grouped together in ρ , and the integer

ambiguities at each frequency are absorbed into bias β . The code is recursively smoothed with the carrier by use of a filter given by [Misra and Enge, 2006]:

$$\hat{P}_i^q = \frac{1}{k} P_i^q + \frac{k-1}{k} \hat{P}_{i-1}^q + \frac{k-1}{k} (L_i^q - L_{i-1}^q) \quad (43)$$

[51] The value of k is equal to the product of the filter time constant and the data sample rate (in our case, 10 s). The smoothed estimate for the i th epoch is $\hat{P}_i^q$. It is an update of the previous estimate at epoch $i-1$ with a weighted combination of the current first-order-iono-free code observables P_i^q and the difference between the current and most recent carrier observables, L_i^q and L_{i-1}^q . The error is the difference between the estimate $\hat{P}_i^q$, and the range ρ_i :

$$\epsilon_i = \hat{P}_i^q - \rho_i \quad (44)$$

$$\begin{aligned} &= \frac{1}{k} P_i^q + \frac{k-1}{k} L_i^q - \rho_i \\ &\quad + \frac{k-1}{k} (\hat{P}_{i-1}^q - L_{i-1}^q) \end{aligned} \quad (45)$$

$$\begin{aligned} &= \frac{1}{k} P_i^q + \frac{k-1}{k} L_i^q - \rho_i \\ &\quad + \frac{k-1}{k} (\hat{P}_{i-1}^q - \rho_{i-1} - \beta) \\ &\quad + \frac{k-1}{k} \left(-\frac{s_{i-1}}{2f_1 f_2 (f_1 + f_2)} - \frac{r_{i-1}}{3f_1^2 f_2^2} \right) \end{aligned} \quad (46)$$

$$\begin{aligned} &= \frac{1}{k} P_i^q + \frac{k-1}{k} L_i^q - \rho_i \\ &\quad + \frac{k-1}{k} (\epsilon_{i-1} - \beta) \\ &\quad + \frac{k-1}{k} \left(-\frac{s_{i-1}}{2f_1 f_2 (f_1 + f_2)} - \frac{r_{i-1}}{3f_1^2 f_2^2} \right) \end{aligned} \quad (47)$$

[52] Expanding the expressions for the i th epoch:

$$\begin{aligned} \epsilon_i &= \frac{1}{k} \left(\rho_i - \frac{s_i}{f_1 f_2 (f_1 + f_2)} - \frac{r_i}{f_1^2 f_2^2} \right) - \rho_i \\ &\quad + \frac{k-1}{k} \left(\rho_i + \frac{s_i}{2f_1 f_2 (f_1 + f_2)} + \frac{r_i}{3f_1^2 f_2^2} \right) \\ &\quad + \frac{k-1}{k} (\beta + \epsilon_{i-1} - \beta) \\ &\quad + \frac{k-1}{k} \left(-\frac{s_{i-1}}{2f_1 f_2 (f_1 + f_2)} - \frac{r_{i-1}}{3f_1^2 f_2^2} \right) \end{aligned} \quad (48)$$

$$\begin{aligned} &= \frac{1}{k} \left(-\frac{s_i}{f_1 f_2 (f_1 + f_2)} - \frac{r_i}{f_1^2 f_2^2} \right) \\ &\quad + \frac{k-1}{k} \left(\frac{s_i - s_{i-1}}{2f_1 f_2 (f_1 + f_2)} + \frac{r_i - r_{i-1}}{3f_1^2 f_2^2} \right) \\ &\quad + \frac{k-1}{k} \epsilon_{i-1} \end{aligned} \quad (49)$$

[53] Equation (49) shows that errors due to higher-order terms arising in the smoothed estimate happen due to (1) large higher-order group errors, and (2) large rates of change in the higher-order phase errors. To examine the effect of each of these, we select lines of sight from the active day that exhibited large magnitudes of group errors and phase error rates.

[54] Figure 8 illustrates the effect of higher-order group errors in the carrier smoothed estimate error. On 29 October 2003, the higher-order group errors on the line of sight (LOS) between the WAAS station in Fort Worth, Texas, and GPS satellite (SVN) 32 built up to nearly 20 cm as the satellite rose and set in the sky. This station-satellite pair produced some of the highest errors, seen at low elevation, shown as the points in the upper left region of Figure 4. Figure 8a shows the higher-order group errors as a function of UTC hour. The y-axis scale ranges from 0 to 20 cm. Figure 8b shows the rate of change of the phase error versus time. Figures 8a and 8b correspond to the terms in the first and second sets of parentheses in equation (49), respectively. Figure 8c shows the smoothing error from these terms, given by equation (44), for two different time constants that bound a range of filter time constants for smoothing multipath. The dashed curve shows the error that accumulates for a filter time constant of 100 s. The 100-s filter tracks the group errors in Figure 8a, reaching nearly 18 cm of error in the smoothed estimate. For a much longer time constant of 2 hours, the errors over time are significantly smaller, at only a couple centimeters. A longer time constant de-weights the first-order-free group observable, and thus the errors associated with it, via the $\frac{1}{k}$ term in (49).

[55] Rates of change in the first-order-free phase observable also contribute to the smoothing error. To examine the impact of these we choose a LOS between the WAAS station in Washington, D.C., and SVN 46. Walter *et al.* [2004] showed this station-satellite pair exhibited extremely high rates of change in the first-order ionosphere group delay. Since the higher-order errors are proportional to the first-order errors, as we found earlier, high rates of change in the first-order error imply high rates of changes in the higher-order errors. Figure 9 shows the smoothing errors for this LOS. Figure 9a shows the higher-order group error as a function of UTC hour. The scale ranges from 0 to 20 cm. The rate of change of the higher-order phase error is

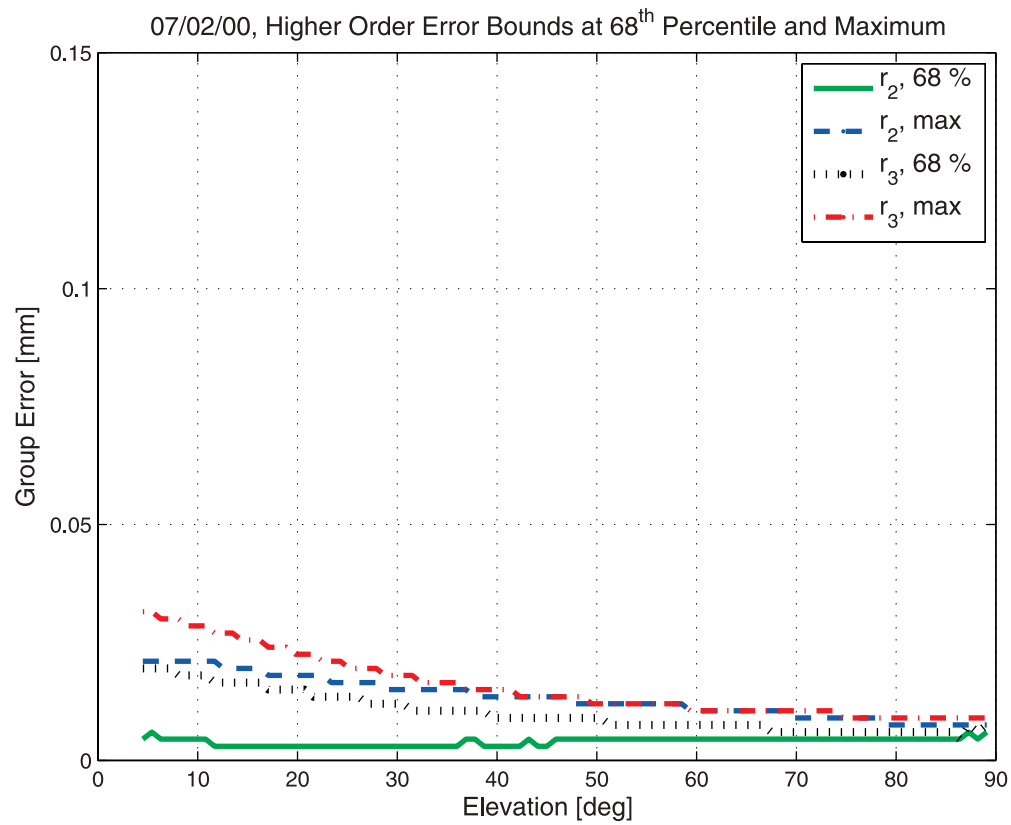


Figure 5. Group P error bounds versus elevation. Third-order $r_2 \sim N_e(B \cos \theta)^2/f^4$ and $r_3 \sim N_e B^2/f^4$ at 68th percentile and maximum value. Quiet day. Y-axis scale: 0 – 0.15 mm.

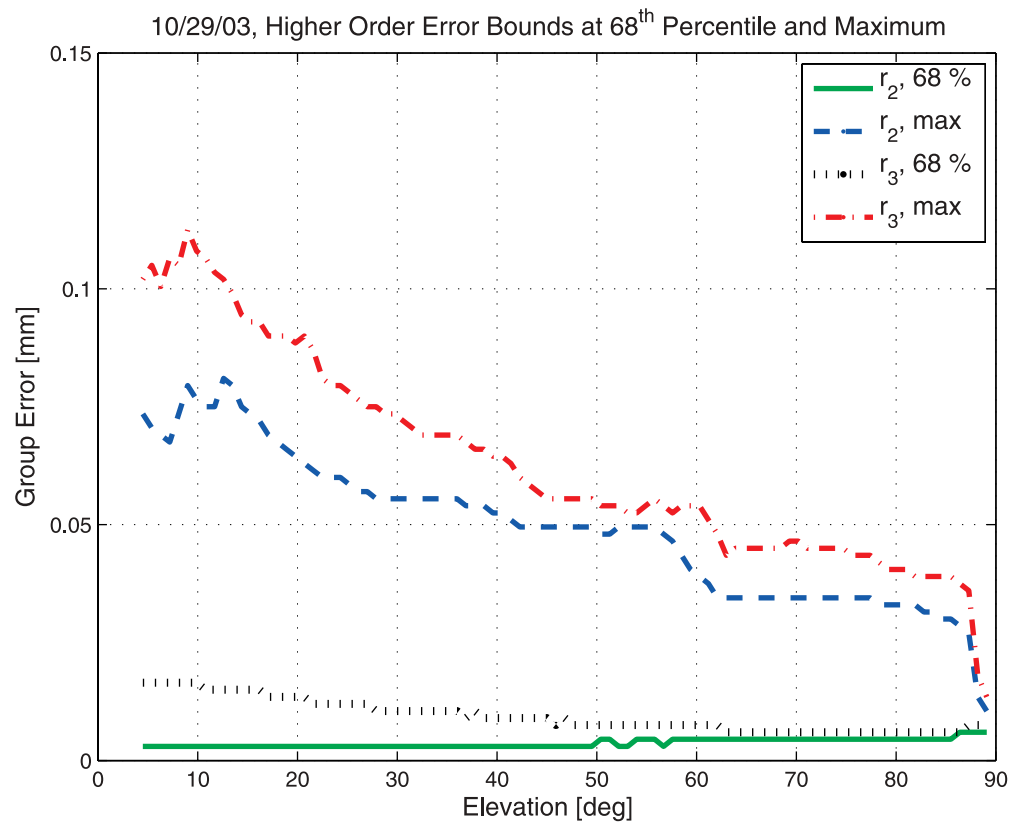


Figure 6. Group P error bounds versus elevation. Third-order $r_2 \sim N_e(B \cos \theta)^2/f^4$ and $r_3 \sim N_e B^2/f^4$ at 68th percentile and maximum value. Stormy day. Y-axis scale: 0 – 0.15 mm.

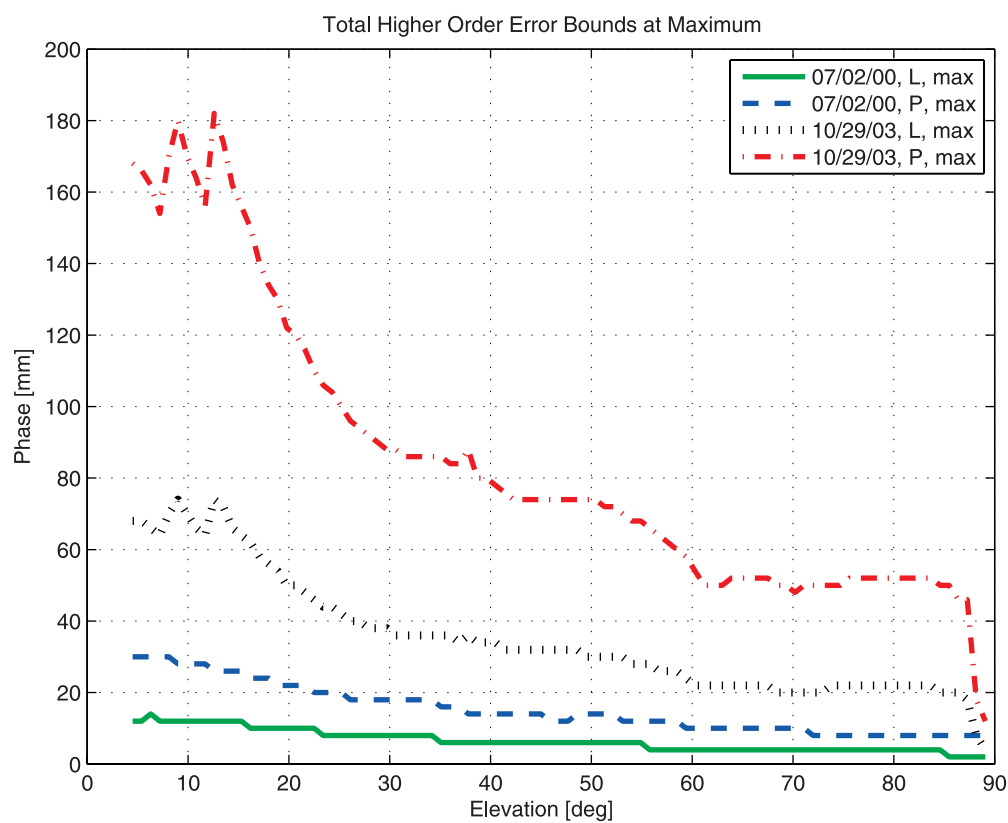


Figure 7. Total higher-order phase error and group error maximum bounds, quiet day and stormy day. Y-axis scale: 0 – 200 mm.

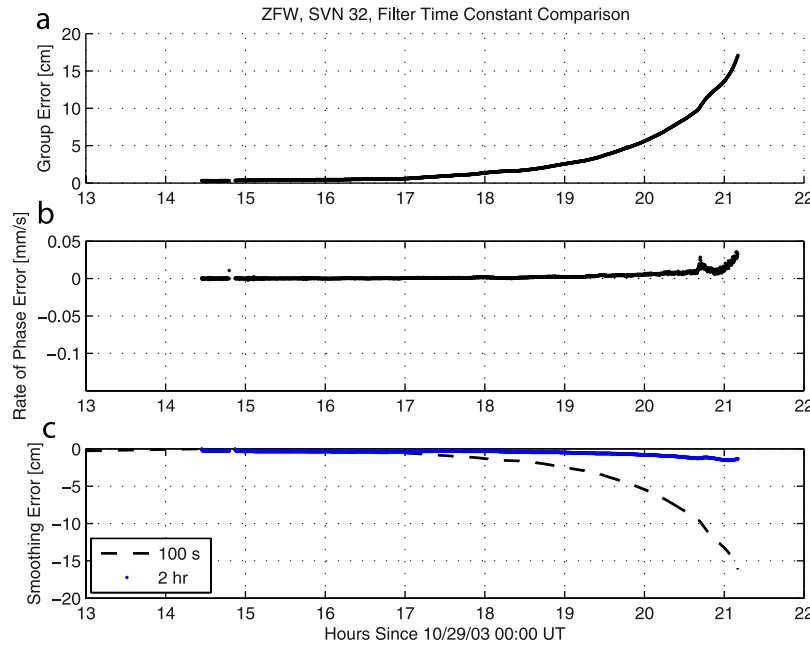


Figure 8. For a line of sight with large higher-order group errors: (a) Higher-order group errors versus time; (b) rate of higher-order phase error versus time; (c) carrier smoothing error of the first-order-ionosphere-free observable for a 100-s time constant (blue) and a 2-hour time constant (green).

shown on Figure 9b. The largest magnitude phase error rate is 0.1 mm/s. Finally the smoothing error, given by (44), is plotted versus UTC hour for two time constants on Figure 9c. The smoothing error for a 100-s filter is shown as a dashed line and for a 2-hour filter as a sequence of dots.

[56] The group error in Figure 9a starts at its highest values, so both filters have almost 4 cm error in the earliest moments. The magnitude of the 100-s filter error then decreases rapidly, tracking the group errors of Figure 9a. The 2-hour filter decreases much more slowly. In fact, the 2-hour filter error magnitude does not reach steady state before a data outage at 23:40 UT causes the filter to reset and begin smoothing a new track. In both cases the anomaly in the ionosphere group error from 20:45 to 21:00 UT is filtered in and appears in the result. For the 100-s time constant (Figure 9c, dashed curve), the smoothing error basically tracks the group error of Figure 9a but is opposite in sign.

[57] In the case of the 2-hour filter result, the anomaly enters the filtering process before a full 2 hours has even elapsed. In our implementation, the instantaneous filter constant $t_{filter,i}$ at the i th epoch equals the minimum of either the target filter constant (e.g. $t_{filter_final} = 2$ hours) or the time elapsed since the first epoch:

$$t_{filter,i} = \min(t_{filter_final}, t_i - t_0) \quad (50)$$

[58] As a result, the filter constant increases with each epoch as the anomaly is filtered into the estimate. For this reason the group anomaly appears in the smoothed estimate with an amplitude that is scaled by a different value at each epoch, giving the curve shown with dots in Figure 9c a different shape with respect to the original group error shown in Figure 9a. The phase error rate of change does not appear to have a significant impact on the error in either the 100-s or the 2-hour filtering cases.

[59] For the L1–L5 dual-frequency user, the higher-order ionosphere errors due to smoothing are a few percent greater in magnitude than those shown in Figures 8 and 9.

8. Conclusion

[60] We used WAAS supertruth data from the most recent solar maximum to compute the magnitude of each of the ionosphere error terms for the L1–L2 dual-frequency user that appear in the literature, up to the negative fourth power of frequency. We found that the *Bassiri and Hajj* [1993] model, which keeps the third power of frequency and only the fourth power term that is proportional to N_e^2 , is justified because it neglects submillimeter errors. We applied this model to data to show that the higher-order phase errors that remain after using the dual-frequency ionosphere correction model maybe up to 8 cm and group errors may reach

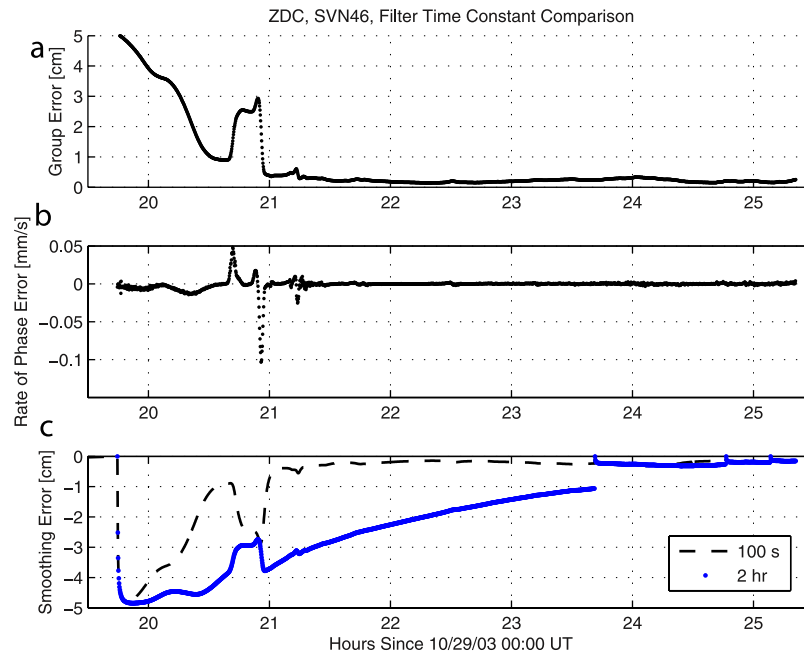


Figure 9. For a line of sight with high phase rate of change: (a) Higher-order group errors versus time; (b) rate of higher-order phase error versus time; (c) carrier smoothing error of the first-order-ionosphere-free observable for a 100-s time constant (blue) and a 2-hour time constant (green).

up to 18 cm at low elevation on days of extreme ionospheric activity. These values are 0.06% and 0.16%, respectively, of the first-order error at L1 frequency. Klobuchar [1996] pointed out that the commonly used dual-frequency model is typically accurate to 0.1%. Our estimate of the higher-order group errors for the active day slightly exceeds those of the simulations in the literature. We attribute this difference to the fact that, in computing the magnitude of the higher-order errors, we deliberately chose the necessary simplifications such that they bounded the errors that may occur. The instantaneous errors for the L1–L5 dual-frequency user will be 6–9% larger.

[61] We showed that the higher-order errors appear in carrier smoothing of the first-order-ionosphere-free code observable. The higher-order group errors, and to a lesser extent the higher-order phase error rates, contribute to the smoothing error. For a 100-s time constant, a low-elevation satellite with a 20 cm higher-order group delay will exhibit a 20 cm smoothing error. This can be reduced to a few centimeters with a longer filter time constant, as we show for a 2-hour filter. The higher-order phase error rate showed no significant effects in either the 100-s or 2-hour filter estimates. The higher-order ionosphere errors in the smoothed L1–L5 dual-frequency estimate are a few percent greater than those for the L1–L2 user.

[62] Higher-order ionosphere errors of magnitude up to 20 cm may be bounded by dual-frequency augmentation systems to provide integrity in one of two ways. The higher-order errors may be estimated in methods similar to the one shown in this paper and removed. Alternatively, the entire higher-order error term may be bounded without removal. Whether one of these methods is better suited to wide-area or local-area differential GPS systems bears further investigation in preparation for the additional civilian GPS frequencies planned.

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